

Accessibility Feedback in Mobile Application Reviews: A Dataset of Reviews and Accessibility Guidelines

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ABSTRACT

User reviews represent a valuable source of information for developers and researchers to learn about the challenges users are facing when using mobile applications (apps). However, studies analyzing mobile app reviews from an accessibility perspective are limited. This paper aims to better understand how accessibility issues are expressed in reviews to reveal potential associations with accessibility guidelines. To this end, we categorized 1,200 accessibility reviews using accessibility categories from previous studies and the W3C Accessibility Guidelines (WCAG 2.1). Our results indicate that accessibility user reviews contain accessibility-specific information. This information, while expressed by users in terms that may differ from the technical terms, were still indicative of specific accessibility guideline violations. Our new dataset can potentially be used by app developers to better understand users' accessibility experience.

KEYWORDS

accessibility, reviews, mobile, apps, guidelines, dataset

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1 INTRODUCTION AND BACKGROUND

In today's landscape, mobile devices have become ubiquitous. The utility they bring to everyday life cannot be understated, being high essential in some disciplines. As such, software developed for these platforms is faced with the challenge of catering to a wide variety of users from all walks of life. One way in which developers can work

towards achieving this goal is by creating accessible apps; in other words, apps that are easy-to-use and welcoming to as many people as possible regardless of physical or cognitive impairment. The lack of accessibility cuts off a percentage of the population from using mobile apps comfortably [4, 31]. With approximately 15% of the world's population living with some form of disability [40], apps that target the largest audience possible cannot underestimate the importance of accessibility.

Typically, there are two main ways to evaluate the accessibility of mobile apps. The first approach is evaluating the mobile app against a checklist, i.e., accessibility guidelines, either by manual inspection conducted by accessibility experts [33] or using automated tools [31, 35]. This type of evaluation is sometimes complemented by a disability simulation software [8], albeit disability simulations do not lead to accurate results [38]. The second approach is user-centered evaluation where end users test the software [15]. Another new approach that can inform us about the accessibility status of apps is examining user reviews. Users can leave feedback about apps by writing reviews (e.g., on Apple's App Store and Google's Play Store), and these reviews can be helpful in revealing the accessibility status of the app as perceived by the end user.

There are several studies exploring the potential of user feedback in user reviews [9, 12, 24, 25, 27, 37]. Studies such as Pagano et al. [24] and Khalid et al.'s [18], provided an extensive list of the types of user feedback commonly encountered. However, these types are high-level, consisting of categories such as "praise", "helpfulness", "bug report", and "feature request" for Pagano and Maleej, and "functional error", "compatibility", and "interface design" for Khalid et al. Another study done by Lu and Liang, based on the categories defined by the International Organization for Standardization (ISO), identified six types of reviews based on the topic they address: Usability, Reliability, Portability, Performance, Feature Request (i.e., "Things that a system/product should do") and Others [20]. Overall, while classifying user reviews is an explored topic, these studies do not go into enough detail to address classification from the perspective of accessibility. To the best of our knowledge, those studies that do approach classification from an accessibility angle are limited in their scope; focusing on the binary classification of reviews as either related to accessibility or not [3, 12].

In this paper, we further studied the accessibility reviews, initially identified by Eler et al. [12], to determine whether we can obtain more accessibility-specific information from an accessibility

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Table 1: The codebook used to categorize the accessibility reviews, with a review example and count of reviews per each category.

Accessibility Category	Definition	Examples	Sample Quote, Source, and Date	Frequency
Text	Relates to the textual information visually transmitted to the user. The format, size, font, and color of the text, as well as text alternatives (Braille, speech, symbols).	Readability of text, change font size, reflow.	<i>"Please fix the text box sizing when in clue view. It's zooming in and you can't see all the letters as soon as it's bigger than 4 characters long."</i> com.totsp.crossword.shortyz (2015)	142
Audio	Relates to the volume, control, clarity, or accessibility of auditory information. Alternative to audio exists (e.g. transcript).	Volume too high, audio plays automatically, ability to pause audio.	<i>"Some audiobooks are very quiet and I can't increase volume. :(Must change for another app I think."</i> de.ph1b.audiobook (2017)	94
Video	Audiovisual controls, and content are accessible (e.g. audio description, subtitles).	Subtitles, video plays automatically, zooming into, navigate video.	<i>"Hey dev - can you add zooming and slow motion to the video?"</i> com.simplerobiletools.gallery.pro (2019)	49
UI Elements	Relates to the visual characteristics of the app's user interface, including images. Elements should be clearly labeled, colored, and positioned.	Color scheme, elements' (buttons, links, etc.) size.	<i>"Signs and numbers ought to have a black outline/border. Any white in your background image makes characters invisible."</i> com.android2.calculator3 (2016)	407
User Control	How the user interacts with the app's user interface, and adhering to system settings, e.g., font, color, and orientation preferences.	Controls, e.g., volume control, enough time to interact with controls.	<i>"Control is a mess compared to any other video player... Specially seeking volume next/previous etc."</i> org.xbmc.kodi (2018)	65
Flexibility and Efficiency	Customization the app allows the users to and the efficiency in completing tasks (least number of steps).	Options, customization, how quick or easy to use the application is.	<i>"Smooth and great personalization options."</i> com.moez.QKSMS (2015)	192
Recognition rather than Recall	All the information needed to complete a task are available in the same screen, without scrolling or revealing hidden information.	No need to scroll or memorize information.	<i>"The song and artist name are almost completely hidden. I can't see the name of the song being played."</i> com.kabouzeid.gramophone (2017)	34
Gestures	How the user interacts with the app from the point of view of the user's physical movements.	Hand movement, perform action with a gesture.	<i>"Good app. Would like pinch and zoom though."</i> net.kervala.comicsreader (2014)	56
System Visibility	The app operates and reacts in ways the user predicts and perceives.	Responsive, unexpected result.	<i>"Keep hang when I browse subtitle."</i> org.xbmc.kodi (2019)	44
Error Prevention	The app helps the user avoid and correct mistakes.	Indicating and correcting wrong input, confirm actions.	<i>"Typing is [too difficult]-replaces words with periods."</i> com.danvelazco.fbwrapper (2015)	11
Tangible Interaction	Relates to physical feedback or peripheral (e.g. physical keyboards).	Haptic, compatible with a peripheral.	<i>"Can you add haptic feedback to typing?"</i> org.pocketworkstation.pckeyboard (2019)	27
Others	The review relates to accessibility but doesn't fit in any of the categories above.	Accessibility topic that does not match the previous categories.	<i>"Warning to anyone with epilepsy or neurological conditions - may not have desired effect! Don't have epilepsy but do have RSD and now in a lot more pain."</i> com.ihunda.android.binauralbeat (2015)	25

review. This study consists of two parts. First, we performed keyword extraction on the reviews to determine relevant features of accessibility categories. Second, a manual classification of a dataset consisting of accessibility reviews, where each review is categorized into one of 12 different categories as well as World Wide Web Consortium (W3C) Web Content Accessibility Guidelines (WCAG). The WCAG provides mobile accessibility guidelines equipped with thorough documentation and testing criteria [39]. The contributions of our work are identifying the types of accessibility feedback reported in reviews, the frequency of those reports, and noting where accessibility guidelines could be expanded to reflect user language and general usability techniques (e.g., dark mode, customization). Our dataset is available on github¹ and can prove useful for further machine learning and HCI applications.

2 METHODOLOGY

2.1 Preparing the Dataset

Our dataset consists of the reviews categorized as accessibility related by Eler et al. [12]. These reviews were acquired from the FDroid app repository², a catalog of Free and Open Source Software apps for Android. The reviews were collected from 701 apps – excluding apps under the category of “Theming and System” and apps that are not available in the Google Play Store. To our knowledge, this is the only available dataset that contains accessibility reviews and was used in a follow-up study [3]. With a list of keywords from the BBC guidelines [6], Eler et al. filtered out non-accessibility reviews. On the second pass, an author manually checked the remaining accessibility related reviews to ensure there were no false positives [12]. Out of the total 214,053 reviews, only 2,663 reviews from 276 apps were categorized as accessibility related by Eler et al. [12]. Of these accessibility reviews, 1,200 were randomly chosen for our analysis and categorization.

2.2 Automatic Keyword Extraction

Automatic keyword extraction is an approach that aims to identify the terms that represent the content of a text document [30]. In order to identify recurrent words observed in the accessibility reviews, we performed automatic keyword extraction on the randomly selected 1,200 accessibility reviews. We used Microsoft Azure’s Text Analytics API (v3.0) which is available in their Cognitive Services suite [21]. Specifically, we used the key phrase extraction service the suite provides. We then developed a Python console application to convert the review’s text into a JSON file in order to input it into Azure API and obtain the list of extracted key phrases. Finally, we obtained a set of keywords from these key phrases to determine what topics frequently occurred with certain accessibility categories.

2.3 Qualitative Analysis

We followed a deductive coding approach [14]. Two coders independently labeled the same 1,200 accessibility reviews used in the keyword extraction. Using Ballantyne et al.’s mobile accessibility categories [5] and the Web Content Accessibility Guidelines

(WCAG) 2.1 [39], they labeled the reviews with the corresponding guideline based on the predominant accessibility topic represented. Then, the two coders convened to compare results, discuss, and reach consensus on matching codes.

We chose the Ballantyne et al.’s mobile accessibility guideline categories as ground truth labels for the reviews because they cover different aspects of mobile accessibility as presented in literature [5]. It covers both the BBC guidelines [6], which were used to build the Eler et al. dataset [12], and the WCAG guidelines [39], which are considered as a standard baseline by many researchers and developers [7, 29, 32]. Additionally, labeling the reviews per mobile accessibility guidelines can give us direct insights on the adherence (or the lack of) to specific accessibility guidelines in the reviewed apps. Additionally, we added the label “Others” to capture reviews that may not fit into any of the defined accessibility categories. We also created a codebook to help determine what kind of reviews belong to each accessibility category, see Table 1.

3 RESULTS

3.1 Findings of the Keyword Extraction

The initial keyword extraction revealed 2,784 unique words. However, some of the keywords refer to the same feature with different terms. For example, the terms “dark”, “black”, and “night” all commonly refer to the same or extremely similar features - dark/black/night mode/theme - and are counted as unique words. The full list of keywords can be seen in the repository¹.

Eler et al.’s [12] study focused on keyword-based binary classification, using a list of 222 keywords that included a few variations (e.g., color blind/colour blind) which were all extracted from the BBC accessibility guidelines [6]. To learn how users referred to accessibility compared to accessibility guidelines, we compared our extracted keywords to Eler et al.’s set, where we found that 125 of the keywords matched. Table 3 shows the top 50 notable recurrent keywords that appeared in our keyword extraction. Words that also appear in Eler et al.’s set are marked with an asterisk. Two omissions from Eler et al.’s list are the words ‘theme’ and ‘option’. ‘Option’ ranks first in terms of frequency from our keywords list with other closely related words, such as ‘customization’, present in Eler et al.’s list. ‘Theme’ is often used as an alternative to ‘mode’ in phrases alluding to ‘dark mode’ and similar features. Eler et al. ranks ‘dark mode’ as the most frequent word observed across all accessibility reviews – an observation our results agree with. The absence of the term ‘font’ by itself in Eler et al.’s set is notable as they looked specifically for phrases like ‘change font’ and ‘font size.’ In their repository, which was used in our comparison, only ‘font sizing’ was listed. The use of phrases as keywords may make missing some reviews possible since users may use partial phrases. This spotlights the difficulty of using keyword-based searches, since one single term can be mentioned in multiple ways, even without taking into consideration the users misspelling the words.

By looking into the keywords for each category, we noticed that the user reviews had terms that are different from the technical terms used in guidelines. For example, terms related to ‘dark theme’ and ‘night mode’ are not mentioned in either the WCAG or BBC Guidelines. Instead, these guidelines mostly focus on ensuring the contrast between the elements of a web page or mobile app are

¹ http://github.com/thekindlab/Dataset_Apps_Reviews_by_Accessibility_Guidelines

² Last extracted on May 22nd, 2019

Table 2: Dictionary of codes for WCAG Categories 1. Perceivable, 2. Operable, and 4. Robust used in classifying the accessibility reviews during the qualitative analysis phase. Also includes the frequency reviews were tagged with each success criterion. *Note: The success criterion omitted from this table have less than 10 reviews coded for that respective criterion and thus were deemed trivial.

Success Criterion	Definition	Examples	Frequency
1.2.4 Captions (Live)	All live audio content in synchronized media should have captions.	Captions for real-time presentations	24
1.3.3 Sensory Characteristics	Understanding and operating the user interfaces isn't dependent solely on one or two sensory characteristics.	Haptic feedback, icon design (labeling, positioning, color)	20
1.4.2 Audio Control	Audio should have a pause and/or stop mechanism or be controlled independent of the overall system volume level.	Pause button, stop button, robust volume settings	142
1.4.4 Resize text	Text should be resizable up to two times its original size without assistive technology. This doesn't apply to captions and images of texts.	Zoom mechanism (e.g. pinch-to-zoom), zoom settings	47
1.4.8 Visual Presentation	Allows users to change color of individual components and/or different color themes. Text should be resizable up to two times its original size without assistive technology so users don't need to scroll horizontally to read the text.	Different modes (e.g. dark mode, light mode), zoom mechanism, zoom settings	454
1.4.11 Non-text Contrast	Components of the user interface must have sufficient contrast among adjacent colors.	Dark mode, light mode themes, contrast	260
1.4.13 Content on Hover or Focus	Content that can appear and disappear based on pointer hover or keyboard focus should be dismissable, hoverable, and persistent.	Hovering tooltips, popup sub-menus, information panels	12
2.2.2 Pause, Stop, Hide	Moving, blinking, scrolling, or auto-updating information should have a mechanism to pause, stop, dismiss, and/or change its frequency.	Pause, stop, and restart buttons, limiting flashes to at most five seconds	16
2.4.6 Headings and Labels	Headings and labels are provided for describing topics or purposes.	Clear and concise labels	18
2.5.1 Pointer Gestures	Content that is operable with multipoint or path-based gestures should also be operable with one pointer unless the multipoint or path-based gesture is considered essential.	Having an alternative button(s) for touch gestures	10
2.5.5 Target Size	Anything that utilizes pointer inputs should be at least be sufficient in size (44 by 44 CSS pixels).	Buttons, links	16
2.5.6 Concurrent Input Mechanisms	Unless necessary, content should not restrict the usage of different input mechanisms.	Touch screen, keyboard and mouse, track pad, voice commands, braille displays	17
4.1 Compatible	Content is compatible and customizable with the current and future devices, including assistive technology.	Customization options, compatible across different devices	105

perceivable. Similarly, terms like 'option' and 'customization' do not appear in the context of providing users with alternatives to adjust their experience, if they appear at all. The guidelines focus on how to make specific features accessible without mentioning customizable options for the user.

In summary, the keyword extraction, with its limitations, helped us detect some of the accessibility information in the user review.

However, the keywords alone do not help with associating a user review to a specific guideline. For further investigation, we conducted qualitative analysis and its results are explained in the following section.

Table 3: Top 50 notable keywords in accessibility reviews.

Words with an asterisk appear in both our study and Eler et al.'s [12]				
theme	options	version	settings	browser
dark*	phone	notification	design	music
mode*	night	feature	problem	interface
customization*	color*	ui	issue	small*
volume*	time	widget*	messages	images*
text*	button*	zoom*	keyboard*	white*
option	background*	ability	buttons	notifications
size*	features	video	player	image*
font*	update	control*	light*	issues
screen*	black*	simple	default*	sound

3.2 Categorizing Accessibility in the Reviews

In classifying the reviews, differences in the employed definition of 'accessibility review' and human error resulted in some variation from Eler et al.'s results. Some reviews contained keywords defined by Eler et al., but were not used in an accessibility context (e.g., explaining a bug or factual statement). 4.5% of the reviews in Eler's dataset are not accessibility related and were excluded. The count of reviews in each accessibility category is shown in the last column in Table 1. The top three categories represent 64.66% of all reviews and are the only categories to contain over 100 reviews each. The majority of accessibility reviews were centered around "UI Elements" (35.51%), "Flexibility and Efficiency" (16.75%), and "Text" (16.75%). "User Control" and "Gestures" account for 10.56% of reviews. These two categories relate to how users physically interact with the apps, either from the perspective of the app's interface or the physical motions required from the user. Together, these two categories are the fourth most represented category. This suggests that how users physically interact with apps and input methods is important.

One category of interest is "Tangible Interaction." While the category implies physical interaction, most reviews in this category referred to haptic feedback. Additionally, physical peripherals were discussed but with a naturally low rate for mobile devices. Another important category, "Error Prevention", handles text and data inputs and how errors are handled. If users do not expect error prevention or the app handles it in a ubiquitous way, these features were unlikely to be mentioned. Accessibility reviews that did not match defined accessibility categories were labeled as "Others", making up 2.18% of the reviews. Three reviews in the "Others" category merely mentioned accessibility, making it vague. The majority of the reviews in the "Others" category belonged to the same app (com.ihunda.android.binauralbeat). These reviews focused on the neurological effects the app produced such as epilepsy. This data is available in our repository³.

We used the same 1,200 app reviews selected from Eler et al.'s dataset for categorizing the reviews using the Web Content Accessibility Guidelines (WCAG) 2.1. Table 2 summarizes the most popular success criterion as well as their frequency within the reviews. Most reviews predominantly fell in the "Perceivable" WCAG

category, which ensures that all aspect of the UI is perceivable to all users. The UI of an app is something that everyone will encounter, making reviews for the UI more common. Four hundred and fifty four reviews (38%) were classified under guideline 1.4.8 which addresses the visual presentation of large chunks of text. Another highly referenced accessibility guideline was the 4.1 guideline "Compatible", which is part of the "Robust" WCAG category, refers to how customizable content is across multiple technologies and devices and within the app itself. Users also commented extensively on guideline 1.4.2, which refers to audio control, specifically the ability to pause or stop audio and control the app's volume independently of the system volume. Some WCAG guidelines were not associated with any user reviews such as WCAG 1.3.6 ("Identify Purpose"), guideline 1.4.9 ("Images of Text"), guideline 2.2.5 ("Re-authenticating"), guideline 3.1.4 ("Abbreviations"), guideline 3.1.5 ("Reading Level"), guideline 3.1.6 ("Pronunciation"), and guideline 3.3.5 ("Help").

Our results show that reading manually through the user reviews, we can infer the accessibility topic that users discuss. Moreover, we can detect the specific accessibility guideline that the review is addressing. Some user reviews discussed accessibility topics that are not covered by WCAG 2.1. Finally, some WCAG 2.1 guidelines were not mentioned in any of the user reviews we analyzed.

4 DISCUSSION

4.1 Accessibility Specific Information in User Reviews

The user reviews we studied did not cover all WCAG 2.1 considerations. Likewise, the WCAG 2.1 guidelines do not cover all user concerns we observed in the data. Our findings suggest that some app features may cause health issues for users, especially those with neurological conditions such as epilepsy. We note that the dataset we worked with does not include disability information about the users. However, some users disclosed their disability in their reviews. Accessibility flaws that effect neurodivergent users may negatively impact the users' well-being [11], making accessibility guidelines very important to protecting the health of users. Most accessibility guidelines have little language relating to neurodiversity [13]. However, The WCAG 2.1 guidelines cover photosensitive and motion sensitive users in section 2.3 *Seizures and*

³http://github.com/thekindlab/Dataset_Apps_Reviews_by_Accessibility_Guidelines

Physical Reactions[39] and is one of the best examples of guidelines that do consider neurodivergence. For example, section 2.3.2 *Three Flashes*[39] requires that nothing on display flashes more than three times a second, but it is Level AAA, the highest level of requirement conformance. In other words, it is not the minimum required level of accessibility. Yet an accessibility violation such as this would not only render an app unusable, but can also cause active harm to neurodiverse users. Moreover, many web developers lack awareness about cognitive web accessibility [28]. Given our findings, we recommend:

- (1) Developers consider user reviews as an accessibility evaluation source, but they must consider the fact that user reviews are accessibility representative and not comprehensive.
- (2) Creating a thorough documentation of neurodiversity related guidelines.
- (3) Considering a severity-based priority system in place of or in addition to the A, AA, AAA levels.

4.2 The Interplay between Usability and Accessibility in User Reviews

Our findings showed that mapping users' accessibility feedback to specific accessibility guidelines is feasible. Hence, designers, developers, and researchers can quickly learn about the accessibility status of an app from the user reviews. The design elements of the user interface (UI), e.g., font size, and the properties of UI elements such as the size of a button or the distance between two buttons, were traditionally considered as usability guidelines [1, 22, 23]. Many of these usability guidelines were transferred to the accessibility guidelines in WCAG 2.1 [39] and BBC guidelines [6]. Part of the overlap between general usability and accessibility guidelines can be due to the nature of the inclusive design principles [10], which consider accessibility as requirements that improve usability for all users, with and without disability [16, 36]. Thus, it was not surprising to come across several accessibility reviews that sounded more like usability-related reviews. The accessibility reviews were classified as such because of the high correlation between the text of the review and the text of the accessibility guidelines, and not because they were disability-related. Our results support the argument that some accessibility violations can also act as obstacles for people without disabilities [2]. We recommend that in future research, more exploration should target the **consolidation of usability requirements and accessibility standards**, as that can make mobile apps easy to use by virtually everyone [34]. **Creating a unified list can push technology one more step forward towards inclusion, as developers and designers can dedicate more efforts to "making room" for users with disabilities [17].**

5 LIMITATIONS AND FUTURE WORK

Out of 2,663 accessibility reviews available from Eler et al.'s dataset [12], only 1,200 reviews were categorized. Qualitatively analyzing the entire dataset would have provided a more definitive understanding of how accessibility categories were represented in user reviews. However, 1200 reviews is a significant sample, representing the original set with a confidence interval of 3 and a confidence level of 99%, thus the overall results would not have changed significantly. Another limitation of the study relates to the lack of users'

demographics information. The dataset we used did not contain information about users' abilities and whether they had a disability. Our results are only limited to interpretations concerning the conformance of apps to mobile accessibility guidelines based on user reviews, and can not be extended to generalized claims about the accessibility of said apps for people with disabilities.

In the future, we plan on adding more labels to the dataset, such as bug report or feature request, and investigating how to automate the multi classification process of accessibility reviews. By developing such a classifier, we can support software developers in locating accessibility bugs and building more accessible apps, as the lack of tools and resources [19] is often cited as a reason for developing inaccessible software [26].

6 CONCLUSION

User reviews represent a rich source of information about the accessibility of mobile applications. We created a dataset of accessibility reviews that were classified by the accessibility category and guideline(s) the reviews discuss. We also studied how users expressed their accessibility concerns and how these expressions are related to the technical terminologies in accessibility guidelines. Our empirical findings suggest that end users discuss specific accessibility features using a taxonomy that is different from the technical terms used in accessibility standards or guidelines. Mapping users' taxonomy to accessibility guidelines can allow designers and developers to quickly infer the discussed accessibility guidelines, and improve their app's accessibility. User reviews also prove to be a potential source of user experience feedback that is more readily available than direct user studies. They may also uncover accessibility issues that are not well-addressed in existing guidelines. We argue that identifying high-risk accessibility mistakes should be highly prioritized and clearly conveyed to designers and developers. Researchers can direct such efforts and help with detecting and determining high-risk accessibility flaws. We also suggest combining both accessibility and usability guidelines for the benefit of people with and without disabilities.

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